

LAB 2 ORIENTATION

```
package com.example.orientation;

import android.content.pm.ActivityInfo;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private boolean isPortrait = true;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button btnToggle = findViewById(R.id.btnToggle);

        btnToggle.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                if (isPortrait) {
                    setRequestedOrientation(ActivityInfo.SCREEN_ORIENTATION_LANDSCAPE);
                } else {
                    setRequestedOrientation(ActivityInfo.SCREEN_ORIENTATION_PORTRAIT);
                }
                isPortrait = !isPortrait;
            }
        });
    }
}
```

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center_horizontal"
    android:padding="20dp"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/tv1"
        android:layout_width="wrap_content"
```

```

        android:layout_height="wrap_content"
        android:layout_marginTop="50dp"
        android:text="Screen Orientation"
        android:textSize="24sp"
        android:textStyle="bold"/>

<Button
    android:id="@+id/btnToggle"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="60dp"
    android:text="Click to change Orientation"/>

</LinearLayout>

```

LAB 3 AND 13 LOGIN CREDENTIALS

```

package com.example.lab3;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    EditText logintext, passwordtext;
    Button btnlogin;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Binding UI components from layout
        logintext = (EditText) findViewById(R.id.textusername);
        passwordtext = (EditText) findViewById(R.id.textpassword);
        btnlogin = (Button) findViewById(R.id.btn1);
    }
}

```

```

// Setting up the login button click listener
btnlogin.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        String user = logintext.getText().toString();
        String pass = passwordtext.getText().toString();

        // Validating credentials
        if (user.equals("Francis") && pass.equals("College@123")) {
            Toast.makeText(getApplicationContext(), "Login Successful", Toast.LENGTH_SHORT).show();
        } else {
            Toast.makeText(getApplicationContext(), "Wrong Credentials", Toast.LENGTH_SHORT).show();
        }
    }
});
}
}

```

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/signin"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="28dp"
        android:gravity="center_horizontal"
        android:text="Sign in"
        android:textColor="@color/black"
        android:textSize="25sp"
        android:textStyle="bold" />

    <TextView
        android:id="@+id/username"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_below="@+id/signin"
        android:layout_marginEnd="50dp"
        android:layout_marginStart="50dp"
        android:text="Username"

```

```
android:textColor="@color/black"
android:textSize="20sp" />
```

```
<EditText
    android:id="@+id/textusername"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@+id/username"
    android:layout_marginLeft="50dp"
    android:layout_marginRight="50dp"
    android:layout_marginTop="8dp"
    android:hint="username"
    android:inputType="textPersonName"
    android:minHeight="48dp" />
```

```
<TextView
    android:id="@+id/password"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@+id/textusername"
    android:layout_marginLeft="50dp"
    android:layout_marginRight="50dp"
    android:layout_marginTop="24dp"
    android:text="Password"
    android:textColor="@color/black"
    android:textSize="20sp" />
```

```
<EditText
    android:id="@+id/textpassword"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@+id/password"
    android:layout_marginLeft="50dp"
    android:layout_marginRight="50dp"
    android:layout_marginTop="8dp"
    android:hint="Password"
    android:inputType="textPassword"
    android:minHeight="48dp" />
```

```
<Button
    android:id="@+id/btn1"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@+id/textpassword"
    android:layout_marginLeft="50dp"
    android:layout_marginRight="50dp"
    android:layout_marginTop="40dp"
    android:text="LOGIN"
    android:textSize="24sp" />
```

```
</RelativeLayout>
```

LAB 4 CONTENT PROVIDER

MainActivity.java

```
package com.example.implicit;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.provider.ContactsContract;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Button btnExplicit, btnImplicit, btnContacts;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnExplicit = findViewById(R.id.btnExplicit);
        btnImplicit = findViewById(R.id.btnImplicit);
        btnContacts = findViewById(R.id.btnContacts);

        btnExplicit.setOnClickListener(v -> {
            Intent intent = new Intent(MainActivity.this, SecondActivity.class);
            intent.putExtra("message", "Hello from MainActivity");
            startActivity(intent);
        });

        btnImplicit.setOnClickListener(v -> {
            Intent intent = new Intent(Intent.ACTION_VIEW);
            intent.setData(Uri.parse("https://www.google.com"));
            startActivity(intent);
        });

        btnContacts.setOnClickListener(v -> {
            Intent intent = new Intent(Intent.ACTION_VIEW, ContactsContract.Contacts.CONTENT_URI);
            startActivity(intent);
        });
    }
}
```

SecondActivity.java

```
package com.example.implicit;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class SecondActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main2);
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center_horizontal"
    android:padding="16dp">

    <Button
        android:id="@+id/btnExplicit"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="100dp"
        android:text="Explicit" />

    <Button
        android:id="@+id/btnImplicit"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="50dp"
        android:text="Implicit" />

    <Button
        android:id="@+id/btnContacts"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="50dp"
        android:text="Content provider" />

</LinearLayout>
```

activity_main2.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello, I have been explicitly called"
        android:textSize="18sp" />

</RelativeLayout>
```

LAB 5 OPENING SCREEN

MAIN ACTIVITY

```
package com.example.prog5;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

Splash_screen.java

```
package com.example.prog5;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.os.Handler;
```

```

public class Splash_screen extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_slplash_screen);

        new Handler().postDelayed(new Runnable() {
            @Override
            public void run() {
                Intent intent = new Intent(Splash_screen.this, MainActivity.class);
                startActivity(intent);
                finish();
            }
        }, 6000);
    }
}

```

Activity main xml

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@drawable/images"
    tools:context=".Splash_screen">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="350dp"
        android:text="@string/displaytext"
        android:textSize="50sp"
        android:textStyle="bold" />

</RelativeLayout>

```

activity_slplash_screen.xml

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@drawable/images2"
    tools:context=".Splash_screen">

```

</RelativeLayout>

LAB 6 UI WITH ALL VIEWS

```
package com.example.views;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.CompoundButton;
import android.widget.ImageButton;
import android.widget.RadioGroup;
import android.widget.Toast;
import android.widget.ToggleButton;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    CheckBox exampleCheckBox;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button example = findViewById(R.id.buttonExample);
        exampleCheckBox = findViewById(R.id.checkBoxExample);
        RadioGroup exampleRadioGroup = findViewById(R.id.radioGroupExample);
        ToggleButton exampleToggleButton = findViewById(R.id.toggleButtonExample);
        ImageButton exampleImageButton = findViewById(R.id.imageButtonExample);

        example.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Toast.makeText(getApplicationContext(),
                    "Button clicked",
                    Toast.LENGTH_SHORT).show();
            }
        });

        exampleImageButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
```

```

        Toast.makeText(getApplicationContext(),
            "Calling Image button",
            Toast.LENGTH_SHORT).show();
    }
});

exampleToggleButton.setOnCheckedChangeListener(
    new CompoundButton.OnCheckedChangeListener() {
        @Override
        public void onCheckedChanged(CompoundButton buttonView, boolean isChecked) {
            if (exampleToggleButton.isChecked()) {
                Toast.makeText(getApplicationContext(),
                    "ON Button clicked",
                    Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(getApplicationContext(),
                    "OFF Button clicked",
                    Toast.LENGTH_SHORT).show();
            }
        }
    }
});

exampleCheckBox.setOnCheckedChangeListener(
    new CompoundButton.OnCheckedChangeListener() {
        @Override
        public void onCheckedChanged(CompoundButton buttonView, boolean isChecked) {
            if (isChecked) {
                Toast.makeText(getApplicationContext(),
                    "Checked",
                    Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(getApplicationContext(),
                    "Unchecked",
                    Toast.LENGTH_SHORT).show();
            }
        }
    }
});

exampleRadioGroup.setOnCheckedChangeListener(
    new RadioGroup.OnCheckedChangeListener() {
        @Override
        public void onCheckedChanged(RadioGroup group, int checkedButtonId) {
            if (checkedButtonId == R.id.radioBtn1) {
                Toast.makeText(getApplicationContext(),
                    "Course selected - Arts",
                    Toast.LENGTH_SHORT).show();
            }
            if (checkedButtonId == R.id.radioBtn2) {
                Toast.makeText(getApplicationContext(),
                    "Course selected - Science",
                    Toast.LENGTH_SHORT).show();
            }
        }
    }
});

```

```

        if (checkedButtonId == R.id.radioBtn3) {
            Toast.makeText(getApplicationContext(),
                "Course selected - Commerce",
                Toast.LENGTH_SHORT).show();
        }
    }
});
}
}

```

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="@string/heading"
        android:textSize="24sp"
        android:textStyle="bold"
        android:gravity="center"
        android:layout_marginBottom="16dp"/>

    <Button
        android:id="@+id/buttonExample"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:backgroundTint="#FF5722"
        android:text="Button"
        android:layout_marginTop="8dp"/>

    <CheckBox
        android:id="@+id/checkboxExample"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:text="@string/checkbox_example"/>

    <ToggleButton
        android:id="@+id/toggleButtonExample"

```

```
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginTop="16dp"
android:text="@string/togglebutton_example"
android:textOn="@string/on"
android:textOff="@string/off"/>
```

```
<RadioGroup
    android:id="@+id/radioGroupExample"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="16dp">
```

```
<RadioButton
    android:id="@+id/radioBtn1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:checked="true"
    android:text="@string/radiobutton_1"/>
```

```
<RadioButton
    android:id="@+id/radioBtn2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="@string/radiobutton_2"/>
```

```
<RadioButton
    android:id="@+id/radioBtn3"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="@string/radiobutton_3"/>
```

```
</RadioGroup>
```

```
<ImageButton
    android:id="@+id/imageButtonExample"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_gravity="center_horizontal"
    android:layout_marginTop="24dp"
    android:src="@drawable/nature"
    android:contentDescription="@string/todo"/>
```

```
</LinearLayout>
```

LAB 7 OPT MENU

```
package com.example.lab7;

import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.opt_menu, menu);
        return true;
    }

    @Override
    public boolean onOptionsItemSelected(@NonNull MenuItem item) {
        int item_id = item.getItemId();

        if (item_id == R.id.option1) {
            Toast.makeText(this, "Opening new file", Toast.LENGTH_SHORT).show();
            return true;
        }
        else if (item_id == R.id.option2) {
            Toast.makeText(this, "Saving file", Toast.LENGTH_SHORT).show();
            return true;
        }
        else if (item_id == R.id.option3) {
            Toast.makeText(this, "Searching file", Toast.LENGTH_SHORT).show();
            return true;
        }

        return super.onOptionsItemSelected(item);
    }
}
```

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android"
```

```

xmlns:tools="http://schemas.android.com/tools"
tools:context=".MainActivity">

<item
    android:id="@+id/option1"
    android:title="New" />

<item
    android:id="@+id/option2"
    android:title="Save" />

<item
    android:id="@+id/option3"
    android:title="Search" />

</menu>

```

LAB 8 READ AND WRITE INTO FILE

```

package com.example.lab8;

import android.content.Context;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;
import java.io.FileInputStream;
import java.io.FileOutputStream;
import java.io.IOException;

public class MainActivity extends AppCompatActivity implements View.OnClickListener {

    Button read, write;
    EditText userInput;
    TextView fileContent;
    String filename = "myfile.txt";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}

```

```

read = findViewById(R.id.read_button);
write = findViewById(R.id.write_button);
userInput = findViewById(R.id.userInput);
fileContent = findViewById(R.id.content);

// Set click listeners
read.setOnClickListener(this);
write.setOnClickListener(this);
}

@Override
public void onClick(View view) {
    Button b = (Button) view;
    String b_text = b.getText().toString();

    switch (b_text.toLowerCase()) {
        case "write":
            writeData();
            break;
        case "read":
            readData();
            break;
    }
}

private void writeData() {
    try {
        FileOutputStream fos = openFileOutput(filename, Context.MODE_APPEND);
        String data = userInput.getText().toString();
        fos.write(data.getBytes());
        fos.flush();
        fos.close();

        userInput.setText(""); // Clear input field
        showMessage("Writing to file " + filename + " completed...");
    } catch (IOException e) {
        e.printStackTrace();
    }
}

private void readData() {
    try {
        FileInputStream fin = openFileInput(filename);
        int a;
        StringBuilder temp = new StringBuilder();
        while ((a = fin.read()) != -1) {
            temp.append((char) a);
        }

        fileContent.setText(temp.toString());
        fin.close();
    }
}

```

```

        printMessage("\nReading from file " + filename + " completed...");
    } catch (IOException e) {
        e.printStackTrace();
        fileContent.setText("Error: File not found or empty.");
    }
}

private void printMessage(String message) {
    fileContent.append("\n" + message);
}
}

```

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text=" File Content "
        android:textAlignment="center"
        android:textColor="#000"
        android:textSize="24sp"
        android:textStyle="bold"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintLeft_toLeftOf="parent"
        app:layout_constraintRight_toRightOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.52" />

    <Button
        android:id="@+id/write_button"
        android:layout_width="wrap_content"
        android:layout_height="48dp"
        android:text="Write"
        app:layout_constraintBottom_toTopOf="@+id/read_button"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.9" />

```

```
<Button
    android:id="@+id/read_button"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Read"
    android:layout_marginBottom="16dp"
    app:layout_constraintTop_toBottomOf="@+id/write_button"
    app:layout_constraintBottom_toTopOf="@+id/textView2"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"/>
```

```
<EditText
    android:id="@+id/userInput"
    android:layout_width="wrap_content"
    android:layout_height="50dp"
    android:layout_marginTop="90dp"
    android:ems="10"
    android:inputType="textPersonName"
    android:hint="Enter text"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />
```

```
<TextView
    android:id="@+id/content"
    android:layout_width="0dp"
    android:layout_height="0dp"
    android:layout_margin="20dp"
    android:text=""
    android:textAlignment="center"
    android:textColor="#000000"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/textView2" />
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

LAB 10 SMS

```
package com.example.lab10;

import android.Manifest;
import android.content.pm.PackageManager;
import android.os.Build;
import android.os.Bundle;
import android.telephony.SmsManager;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;

public class MainActivity extends AppCompatActivity {

    private EditText et1, et2;
    private Button btn1;
    private static final int MY_PERMISSIONS_REQUEST_SEND_SMS = 0;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        et1 = findViewById(R.id.editText1);
        et2 = findViewById(R.id.editText2);
        btn1 = findViewById(R.id.btnSendSMS);

        if (ContextCompat.checkSelfPermission(this, Manifest.permission.SEND_SMS)
            != PackageManager.PERMISSION_GRANTED) {
            ActivityCompat.requestPermissions(this,
                new String[]{Manifest.permission.SEND_SMS}, MY_PERMISSIONS_REQUEST_SEND_SMS);
        }

        btn1.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                sendSMSMessage();
            }
        });
    }
}
```

```

protected void sendSMSMessage() {
    String number = et1.getText().toString();
    String message = et2.getText().toString();

    if (number.isEmpty() || message.isEmpty()) {
        Toast.makeText(this, "Please enter both number and message", Toast.LENGTH_SHORT).show();
        return;
    }

    try {
        SmsManager smsManager;
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.M) {
            smsManager = this.getSystemService(SmsManager.class);
        } else {
            smsManager = SmsManager.getDefault();
        }

        smsManager.sendTextMessage(number, null, message, null, null);
        Toast.makeText(getApplicationContext(), "Message sent success", Toast.LENGTH_LONG).show();
    } catch (Exception e) {
        Toast.makeText(getApplicationContext(), "SMS failed, please try again.",
Toast.LENGTH_LONG).show();
        e.printStackTrace();
    }
}
}

```

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="40dp">

    <EditText
        android:id="@+id/editText1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="60dp"
        android:hint="Enter Phone Number"
        android:inputType="phone"
        android:minHeight="48dp" />

    <EditText
        android:id="@+id/editText2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="50dp"
        android:hint="Enter Message"

```

```

        android:inputType="textMultiLine"
        android:minHeight="48dp" />

<Button
    android:id="@+id/btnSendSMS"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="60dp"
    android:text="Send SMS" />

</LinearLayout>

```

LAB 11 EMAIL

```

package com.example.lab11;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Button button;
    EditText sendto, subject, body;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        sendto = findViewById(R.id.et1_emailid);
        subject = findViewById(R.id.et2_emailsub);
        body = findViewById(R.id.et3_emailmsg);
        button = findViewById(R.id.btn_Send);

        button.setOnClickListener(view -> {
            String emailsend = sendto.getText().toString();
            String emailsubject = subject.getText().toString();
            String emailbody = body.getText().toString();

            Intent intent = new Intent(Intent.ACTION_SENDTO);

```

```

        intent.setData(Uri.parse("mailto:"));

        intent.putExtra(Intent.EXTRA_EMAIL, new String[]{emailsend});
        intent.putExtra(Intent.EXTRA_SUBJECT, emailsubject);
        intent.putExtra(Intent.EXTRA_TEXT, emailbody);

        startActivity(Intent.createChooser(intent, "Choose an Email client:"));
    });
}
}

```

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/tv1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="20sp"
        android:text="Send Email"
        android:textStyle="bold" />

    <EditText
        android:id="@+id/et1_emailid"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="30dp"
        android:hint="Enter recipient email id"
        android:inputType="textEmailAddress"
        android:minHeight="48dp" />

    <EditText
        android:id="@+id/et2_emailsub"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:hint="Enter email subject"
        android:inputType="textEmailSubject"
        android:minHeight="48dp" />

```

```

<EditText
    android:id="@+id/et3_emailmsg"
    android:layout_width="match_parent"
    android:layout_height="200dp"
    android:layout_marginTop="30dp"
    android:gravity="top"
    android:hint="Enter your message"
    android:inputType="textMultiLine"
    android:minHeight="48dp" />

<Button
    android:id="@+id/btn_Send"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_gravity="center_horizontal"
    android:layout_marginTop="20dp"
    android:text="Send email!!" />

</LinearLayout>

```

LAB 12 MAPS

```

package com.example.lab12;

import androidx.fragment.app.FragmentActivity;
import android.os.Bundle;
import com.google.android.gms.maps.CameraUpdateFactory;
import com.google.android.gms.maps.GoogleMap;
import com.google.android.gms.maps.OnMapReadyCallback;
import com.google.android.gms.maps.SupportMapFragment;
import com.google.android.gms.maps.model.LatLng;
import com.google.android.gms.maps.model.MarkerOptions;
import com.example.lab12.databinding.ActivityMapsBinding;

public class MapsActivity extends FragmentActivity implements OnMapReadyCallback {

    private GoogleMap mMap;
    private ActivityMapsBinding binding;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        binding = ActivityMapsBinding.inflate(getLayoutInflater());
    }
}

```

```
setContentView(binding.getRoot());

SupportMapFragment mapFragment = (SupportMapFragment) getSupportFragmentManager()
    .findFragmentById(R.id.map);
if (mapFragment != null) {
    mapFragment.getMapAsync(this);
}
}

@Override
public void onMapReady(GoogleMap googleMap) {
    mMap = googleMap;

    LatLng indiaLocation = new LatLng(12, 77);

    mMap.addMarker(new MarkerOptions().position(indiaLocation).title("Marker in INDIA"));
    mMap.moveCamera(CameraUpdateFactory.newLatLngZoom(indiaLocation, 10f));
}
}
```